

System Name: SkyBridge Airways Integrated Flight Operations, Maintenance, and Turnaround Control Platform

Primary Objective:

Provide a unified operational platform for flight dispatch, crew recovery, maintenance event control, turnaround coordination, EFB package synchronization, and regulated operational record retention.

Trust Boundary: Corporate Identity and Core Records Boundary

Contains:

- Active Directory Forest [Process]
- Federated Identity Service [Process]
- Privileged Access Broker [Process]
- Records Retention Vault [Data Store]
- On-Prem Operational Mirror Database [Data Store]
- Safety Investigation Workspace [Process]

Trust Boundary: Cloud Operations Platform Boundary

Contains:

- API Gateway [Process]
- Operations Workflow Service [Process]
- Dispatch Release Service [Process]
- Crew Recovery Engine [Process]
- Maintenance Control Service [Process]
- Turnaround Coordination Service [Process]
- EFB Sync Service [Process]
- Operational Messaging Service [Process]
- Audit and Decision Log [Data Store]
- Operations Data Lake [Data Store]
- Rules and Policy Engine [Process]

Trust Boundary: Airport and Station Operations Boundary

Contains:

- Station Operations Portal [Process]
- Ground Handler Partner API [Process]
- Fueling Status Adapter [Process]
- De-Icing Status Adapter [Process]

Trust Boundary: Aircraft and Crew Edge Boundary

Contains:

- Electronic Flight Bag Client [External Entity]
- ACARS / Satcom Messaging Provider [External Entity]
- Aircraft Health Feed [External Entity]

Trust Boundary: External Aviation Services Boundary

Contains:

- Weather Intelligence Provider [External Entity]
- Flight Plan Optimization Vendor [External Entity]
- Airport CDM Feed [External Entity]
- Civil Aviation Regulatory Exchange [External Entity]
- Border / Crew Clearance Interface [External Entity]
- MRO Vendor System [External Entity]

Trust Boundary: Vendor Support and Security Boundary

Contains:

- Vendor Support Enclave [External Entity]
- SOC Monitoring Platform [External Entity]
- SIEM / Telemetry Pipeline [Process]

Critical Assets:

- Dispatch Release Package
- Aircraft Tail Status
- Maintenance Deferral State
- Return-to-Service Approval
- Crew Legality Status
- Turnaround Readiness State
- EFB Flight Folder
- Audit and Decision Log
- Break-Glass Approval Record

Privileged Workflows:

- Dispatcher finalizes flight release after reviewing weather, aircraft status, crew legality, and MEL constraints.
- Maintenance controller creates engineering events and proposes deferrals.
- Licensed maintenance supervisor approves selected deferrals and return-to-service decisions.
- Crew recovery manager reassigns pairings during disruptions under legality constraints.
- Reliability engineer opens break-glass diagnostics session through Privileged Access Broker.
- Vendor support operator receives time-boxed diagnostic access after airline approval.
- Safety investigator retrieves immutable decision records from Records Retention Vault.

Security Properties:

- Federated Identity Service: MFA required for workforce users; phishing-resistant MFA required for dispatch, maintenance control, and privileged roles
- Privileged Access Broker: just-in-time elevation; time-boxed sessions; approval required for vendor support; session recording enabled
- Audit and Decision Log: append-only application log; replicated to Records Retention Vault
- Records Retention Vault: immutable retention controls; separate admin domain
- EFB Sync Service: signed package manifest required before client synchronization
- Operational Messaging Service: message integrity checks for provider handoff acknowledgements
- Ground Handler Partner API: mTLS for partner integrations; event-level attribution required
- Turnaround Coordination Service: role separation between station updates and maintenance authority
- Dispatch Release Service: dual-condition approval for selected release exceptions
- Maintenance Control Service: licensed-role check for return-to-service and selected MEL deferrals
- Operations Data Lake: analytics-only replica; not authoritative for operational control
- SIEM / Telemetry Pipeline: receives high-value audit events from dispatch, maintenance, crew recovery, and privileged access workflows

Abuse Cases To Consider:

- False turnaround readiness causing unsafe or non-compliant departure pressure
- Manipulated maintenance deferral or return-to-service state
- Crew legality override abuse during disruption recovery
- Wrong-tail or stale dispatch package synchronized to pilot EFB
- Unauthorized vendor diagnostic access during major incident response
- Tampered health-feed or MRO updates causing incorrect maintenance decisions
- Suppressed or altered audit trail for dispatch or maintenance actions
- Abuse of emergency workflows to bypass normal segregation of duties